

Testing Plan

Testing Scope

Testing in Sprint 1 focuses on validating the core functionality of the MVP, including data handling, vehicle and service logic, and basic user interaction through the terminal-based prototype. Because the system uses simulated vehicle data, testing is limited to logical correctness rather than real-world accuracy.

Testing Objectives

- Verify that sample vehicle data is loaded and stored correctly
- Ensure maintenance service recommendations are generated based on mileage rules
- Validate correct interaction flow in the terminal interface
- Identify and document defects early in development

Types of Testing

Unit Testing

Unit testing will focus on individual classes and methods, including:

- Vehicle mileage updates
- Service due calculations
- JSON data creation and loading

Unit tests will be manual or lightweight, as automated testing is planned for future sprints.

Integration Testing

Integration testing will verify interactions between major components, including:

- DataManager loading vehicle data into the system
- InteractionController accessing vehicle and service information

- Correct data flow between classes

System Testing

System testing will be performed by running the prototype end-to-end to ensure:

- Users can view vehicle information
- Maintenance recommendations are displayed correctly
- The application behaves as expected under normal usage

Test Environment

- **Programming Language:** Python
- **Execution Environment:** Local development machines
- **Data Source:** Simulated JSON vehicle data
- **Interface:** Terminal-based user interface

Defect Tracking

Defects identified during testing will be documented using a standardized bug report template. Each report will include a description of the issue, steps to reproduce, expected behavior, and actual behavior. Bug resolution will be prioritized in future sprints.

Roles and Responsibilities

All team members are responsible for testing features they implement. The team lead coordinates testing efforts and ensures that defects are documented and reviewed.

Limitations

Testing in Sprint 1 does not include performance, security, or real vehicle data validation. These areas are planned for later sprints as system functionality expands.